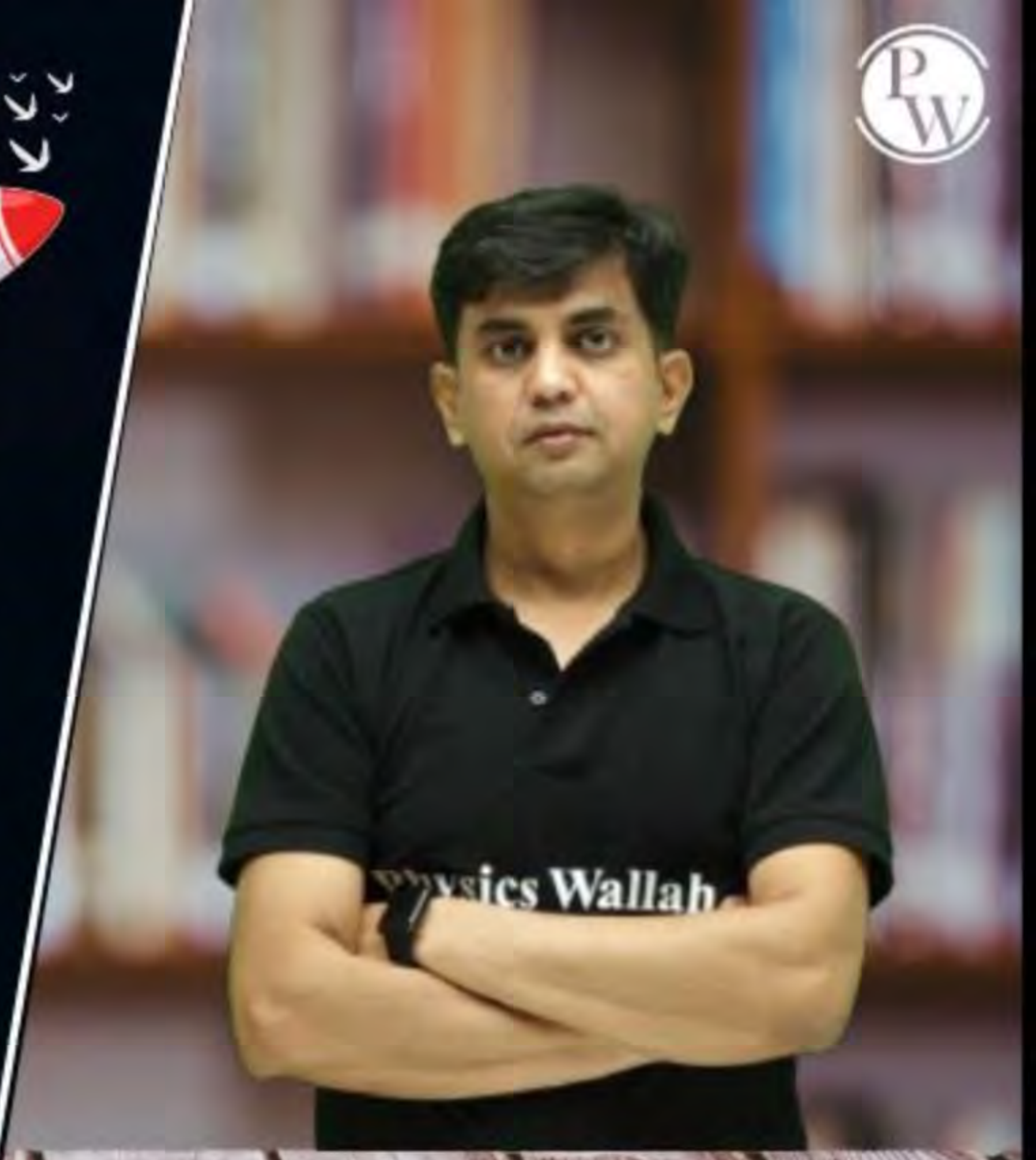
CS & IT

ENGINEERING

Computer Network - 1

Transport Layer

Lecture No. – 10



By - Abhishek Sir

Recap of Previous Lecture

Topic TCP Congestion Control

Topic MSL

Topics to be Covered

Topic TCP Connection Close

ABOUT ME

Hello, I'm Abhishek

- GATE CS AIR - 96
- M.Tech (CS) - IIT Kharagpur
- 12 years of GATE CS teaching experience

Telegram Link : https://t.me/abhisheksirCS_PW

@ABHISHEKSIRCS_PW

[MSQ] [GATE-2023][2 Mark]

#Q. Suppose you are asked to design a new reliable byte-stream transport protocol like TCP. This protocol, named $\boxed{\text{myTCP,}}$ runs over a $\boxed{\text{100 Mbps}}$ network with Round Trip Time of 150 milliseconds and the maximum segment lifetime of 2 minutes. Which of the following is/are valid lengths of the Sequence Number field in the myTCP header?

[RTT << MSL]

(A) ~~30 bits~~

(B) 32 bits

(C) 34 bits

(D) 36 bits

$\boxed{\text{Ans} \geq 31}$

$\boxed{\text{Ans: B, C \& D}}$

Solution :-

Maximum number of bytes can be transmitted in one MSL = MSL * Bandwidth

$= 2\,min * 100\,Mbps = 120\,sec * 10^8\,bits/sec$

$= 12 * 10^9\,bits = \left(\frac{12 * 10^9}{8}\right) bytes$

Minimum number of bits required for sequence number field

= $\lceil \log_2(\text{Max no. of bytes can be transmitted in one MSL}) \rceil$

$= \left\lceil \log_2\left(\frac{12 * 10^9}{8}\right) \right\rceil bits = \lceil 30.48 \rceil\,bits = 31\,bits$

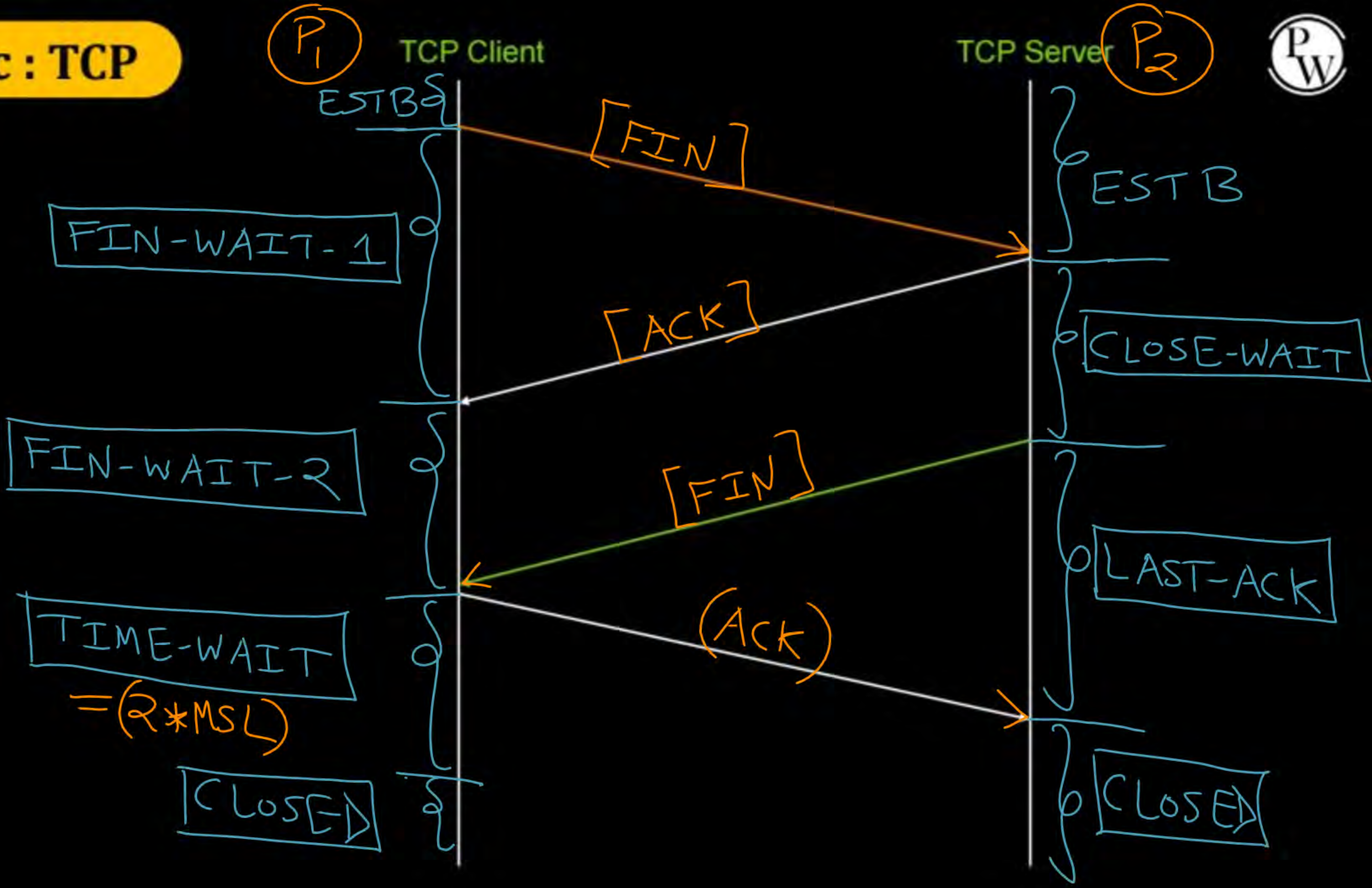
Topic : TCP Connection Close

→ Connection close between TCP client and TCP server

→ 4-way handshake process

→ TCP client or TCP server any one can initiate the connection close request

Topic : TCP



P1
TCP Client
TCP Server
P2
ESTB
[FIN]
ESTB
FIN-WAIT-1
[ACK]
CLOSE-WAIT
FIN-WAIT-2
[FIN]
LAST-ACK
TIME-WAIT
=(2*MSL)
(ACK)
CLOSED
CLOSED

Topic : TCP

P1 TCP Client

TCP Server P2

X

[Client_ISN = 299]

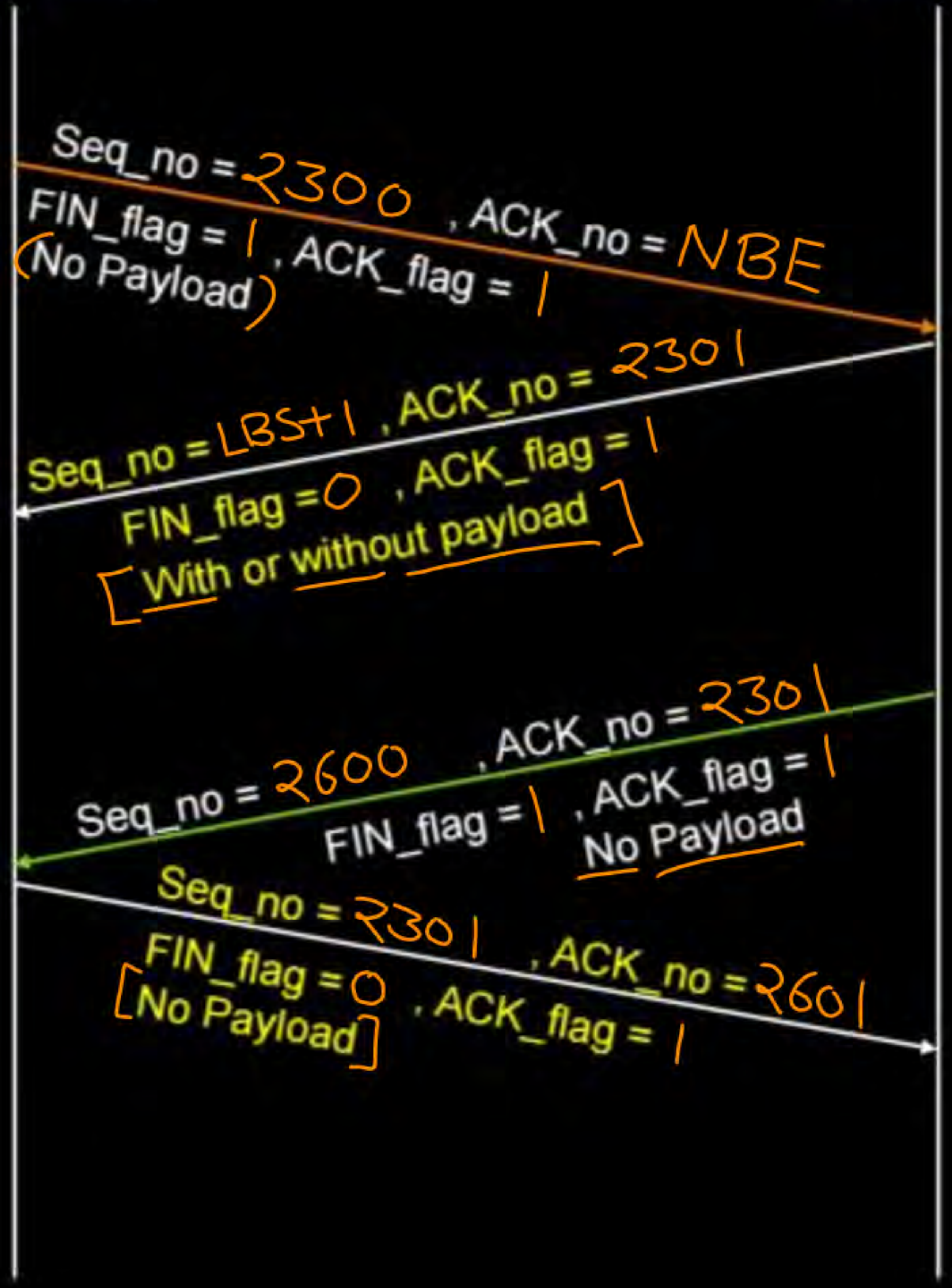
After sending 2000 bytes by client. And all bytes are acked.

First Byte = 300

LastByteSent = 2299

Seq. No. = LBS + 1

Ack No. = Next Byte Expected



Y

Server_ISN = 1599

After sending 1000 bytes by server. And all bytes are acked.

First Byte = 1600

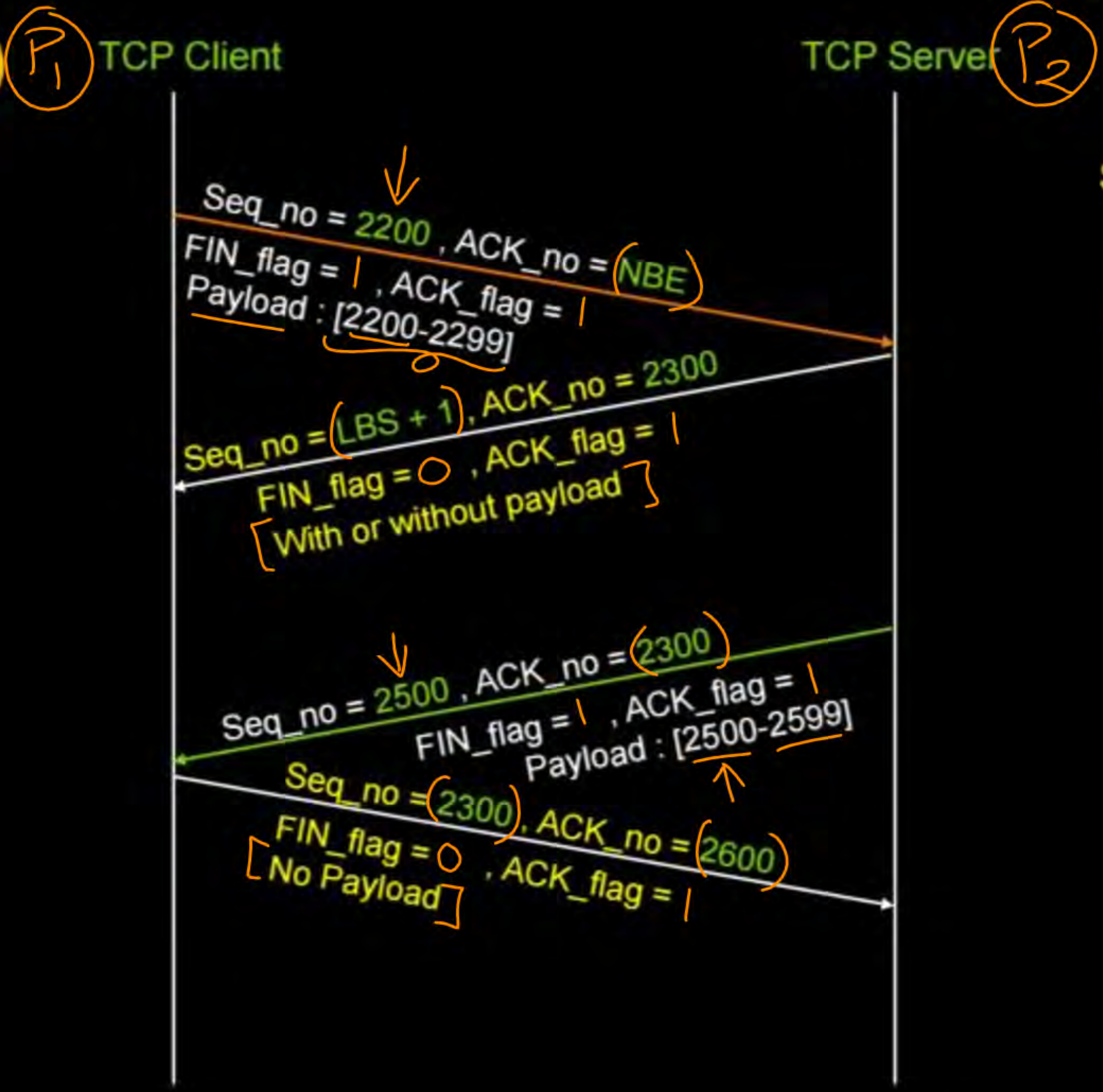
LBS = 2599

Topic : TCP

Client_ISN = 299

MSS = 100 bytes

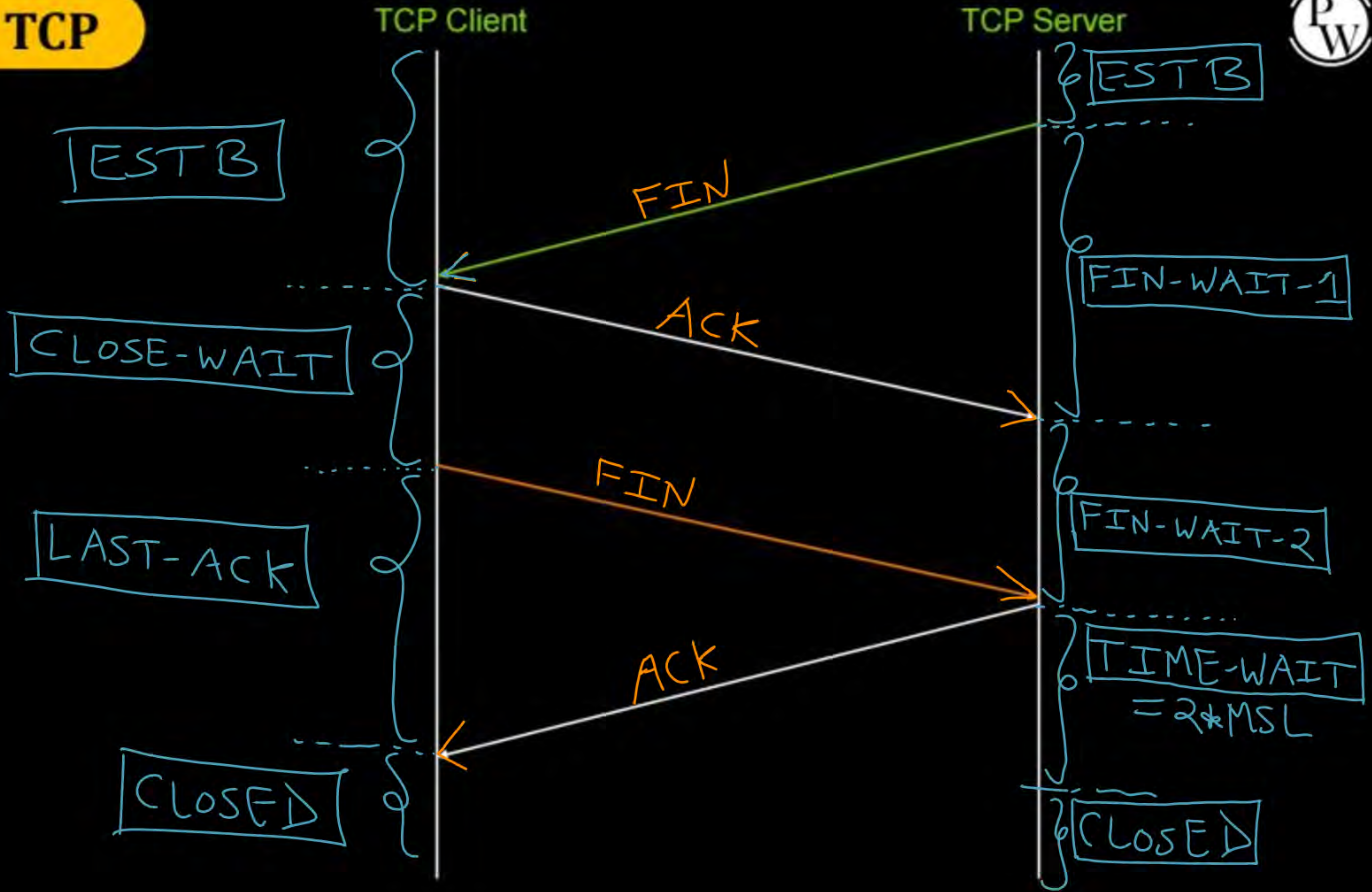
Client sending 2000 bytes.
Starting 1900 bytes are sent and acked.



Server_ISN = 1599

Server sending 1000 bytes.
Starting 900 bytes are sent and acked.

Topic : TCP

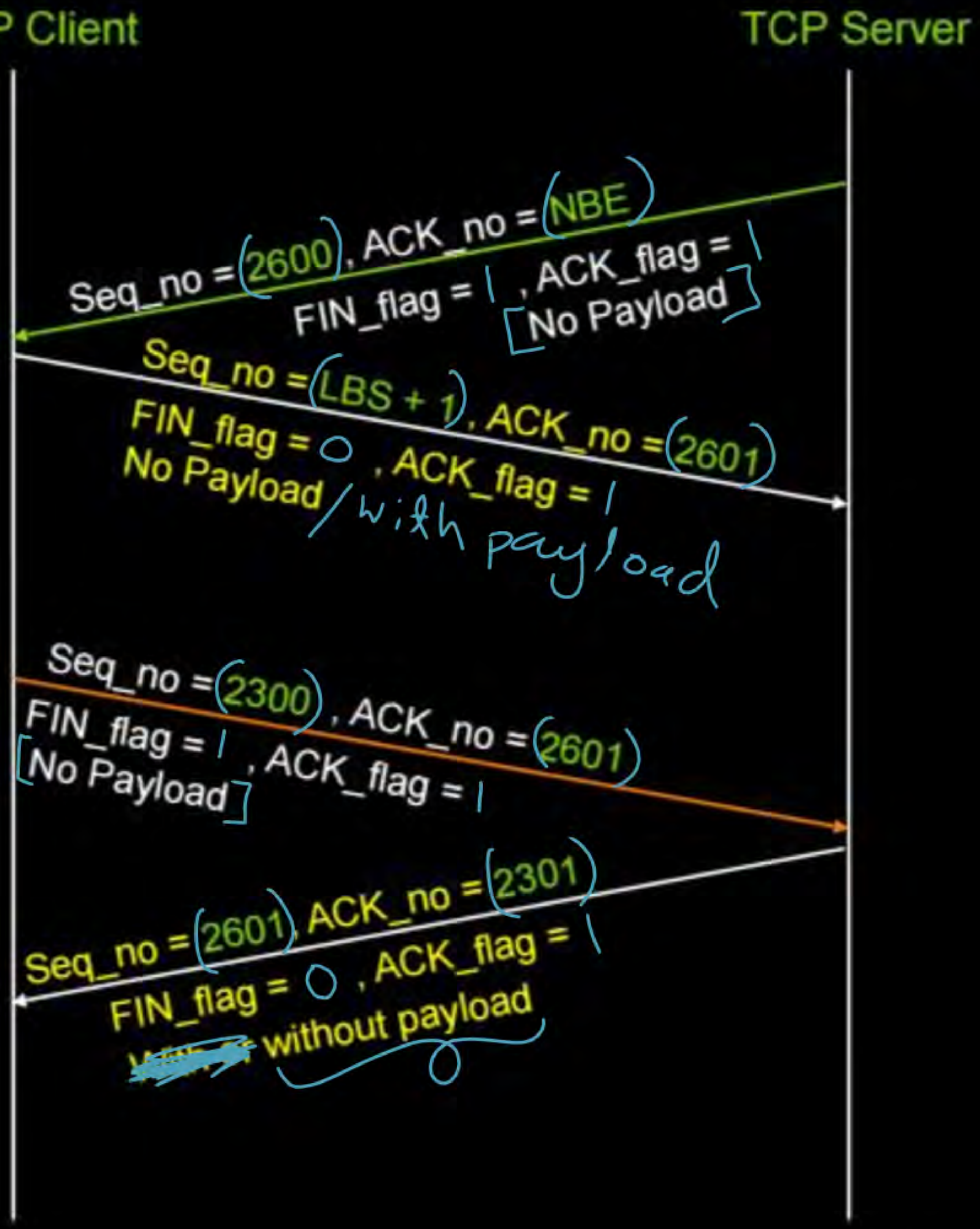


TCP Client
TCP Server
ESTB
ESTB
FIN
FIN-WAIT-1
CLOSE-WAIT
ACK
FIN
FIN-WAIT-2
LAST-ACK
ACK
TIME-WAIT
=2*MSL
CLOSED
CLOSED

Topic : TCP

Client_ISN = 299

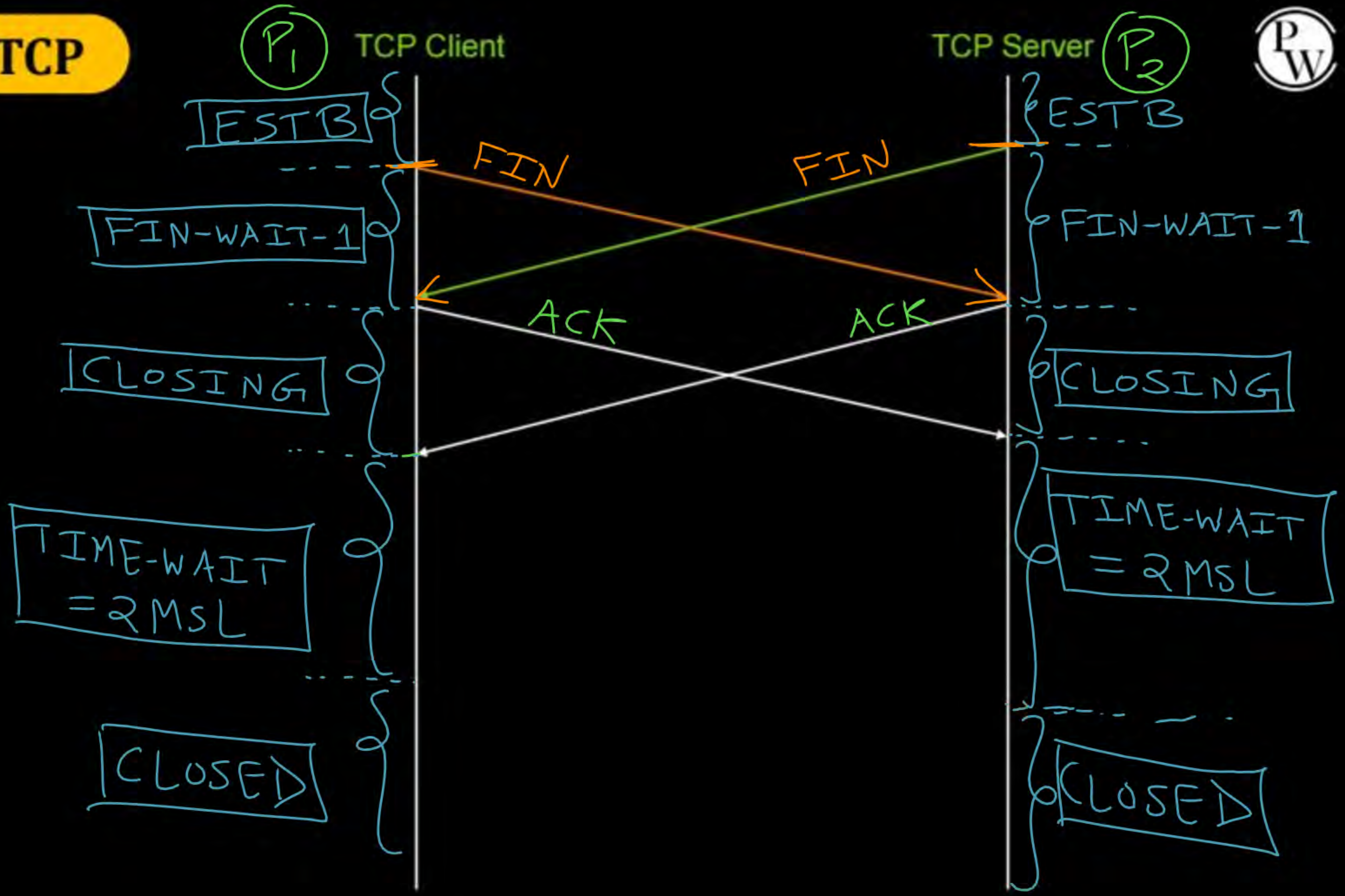
After sending
2000 bytes
by client.
And all bytes
are acked.



Server_ISN = 1599

After sending
1000 bytes
by server.
And all bytes
are acked.

Topic : TCP

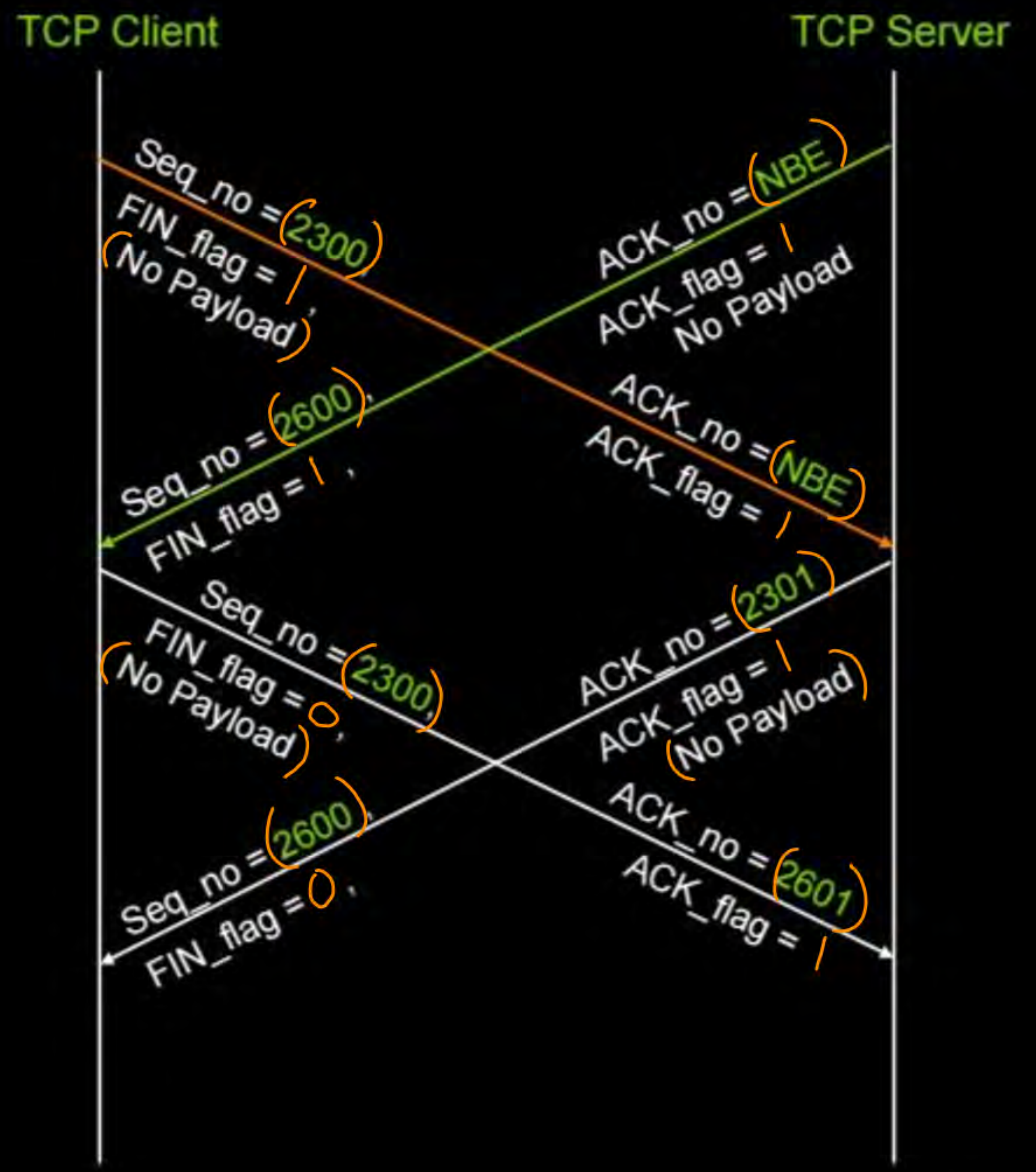


P1
TCP Client
TCP Server
P2
ESTB
ESTB
FIN
FIN
FIN-WAIT-1
FIN-WAIT-1
ACK
ACK
CLOSING
CLOSING
TIME-WAIT
=2MSL
TIME-WAIT
=2MSL
CLOSED
CLOSED

Topic : TCP

Client_ISN = 299

After sending 2000 bytes by client.
And all bytes are acked.



Server_ISN = 1599

After sending 1000 bytes by server.
And all bytes are acked.

[MCQ] (IIT-R) [GATE-2017][2 Mark]

#Q. Consider a TCP client and a TCP server running on two different machines. After completing data transfer, the TCP client calls close to terminate the connection and a FIN segment is sent to the TCP server. Server-side TCP responds by sending an ACK which is received by the client-side TCP. As per the TCP connection state diagram (RFC 793), in which state does the client side TCP connection wait for the FIN from the server-side TCP ?

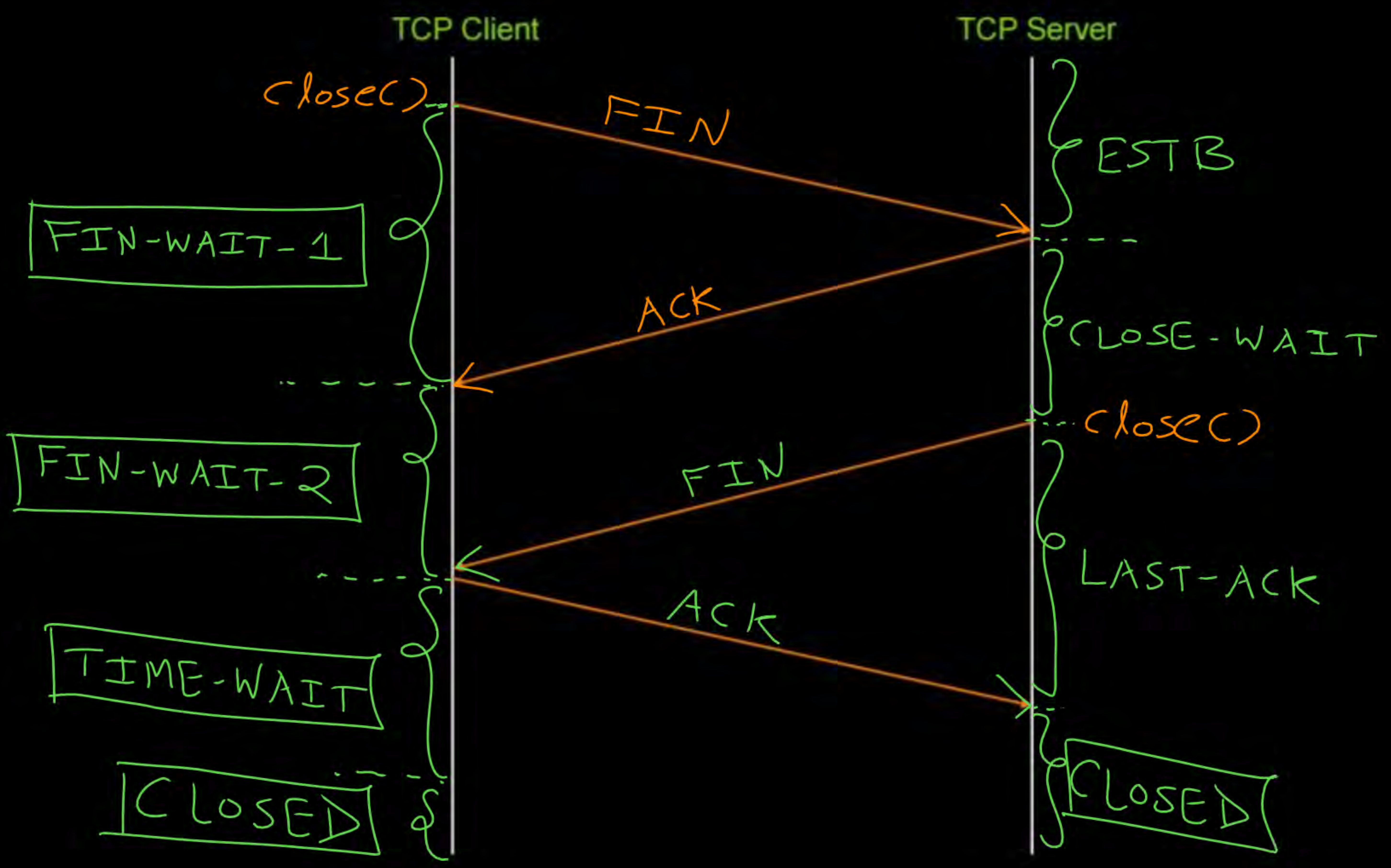
(A) LAST-ACK

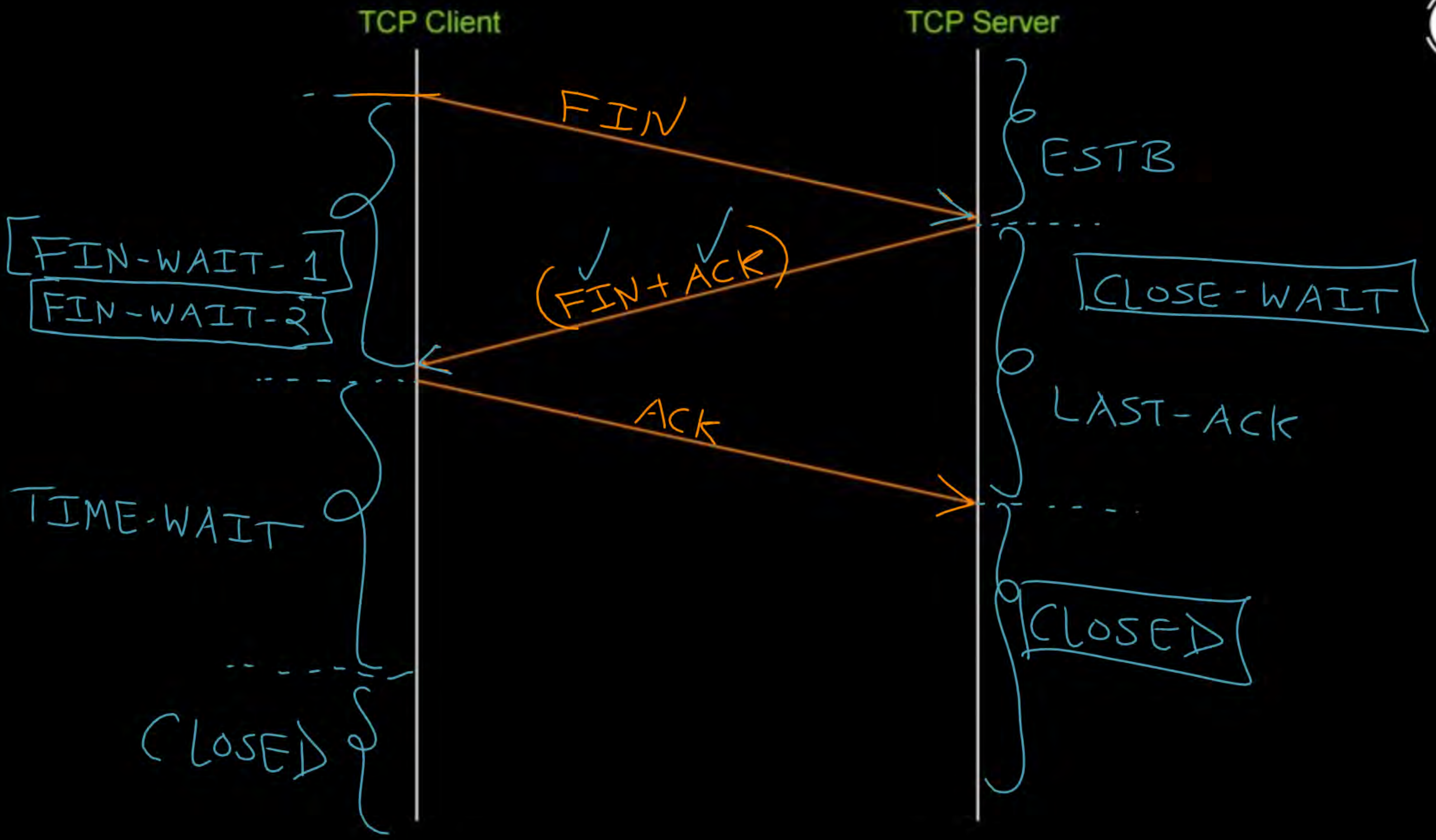
(B) TIME-WAIT

(C) FIN-WAIT-1

(D) FIN-WAIT-2

Ans: D


TCP Client
TCP Server
close()
FIN
ESTB
FIN-WAIT-1
ACK
CLOSE-WAIT
close()
FIN-WAIT-2
FIN
LAST-ACK
ACK
TIME-WAIT
CLOSED
CLOSED

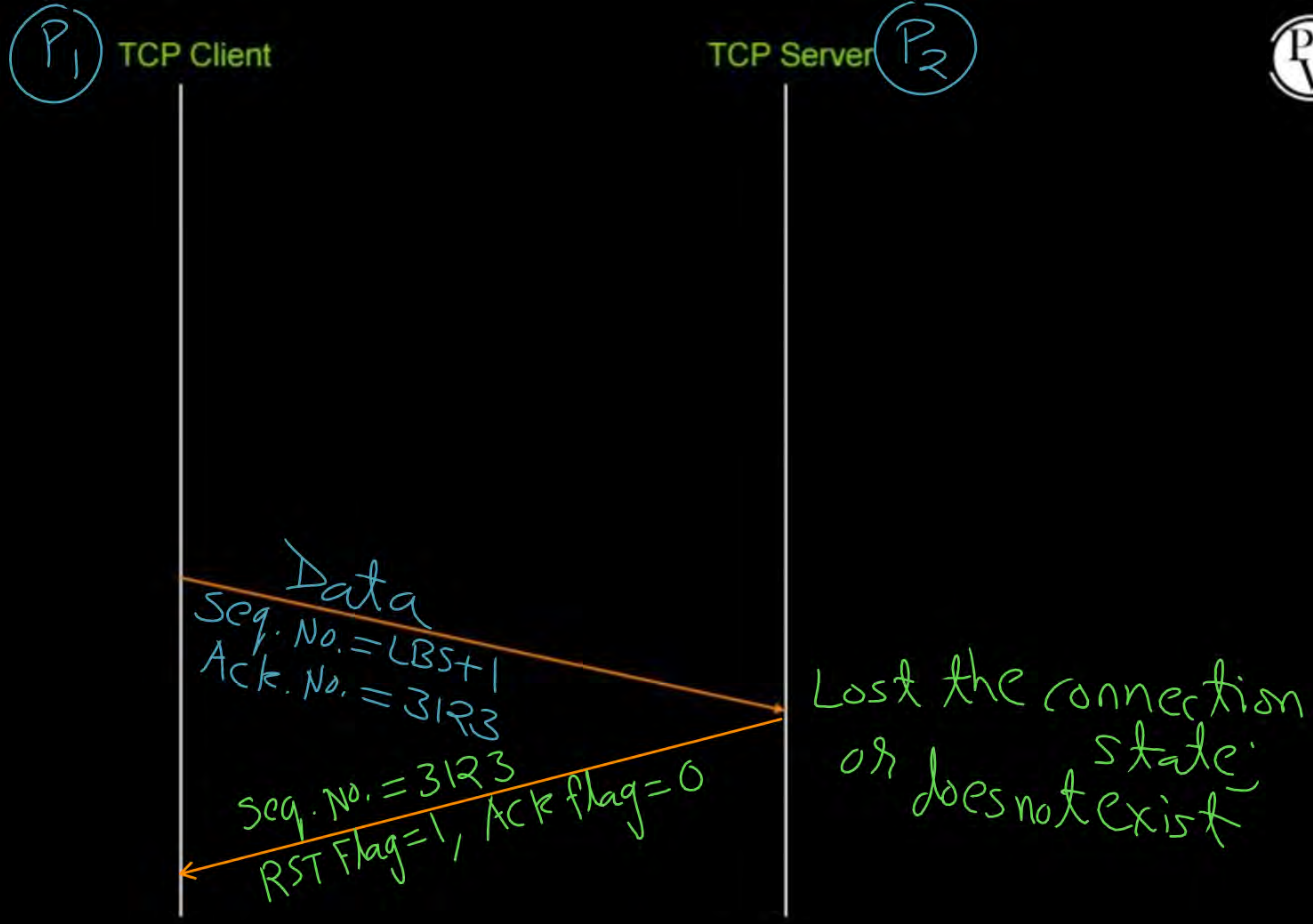

TCP Client
TCP Server
FIN
ESTB
FIN-WAIT-1
FIN-WAIT-2
(FIN + ACK)
CLOSE-WAIT
LAST-ACK
ACK
TIME-WAIT
CLOSED
CLOSED

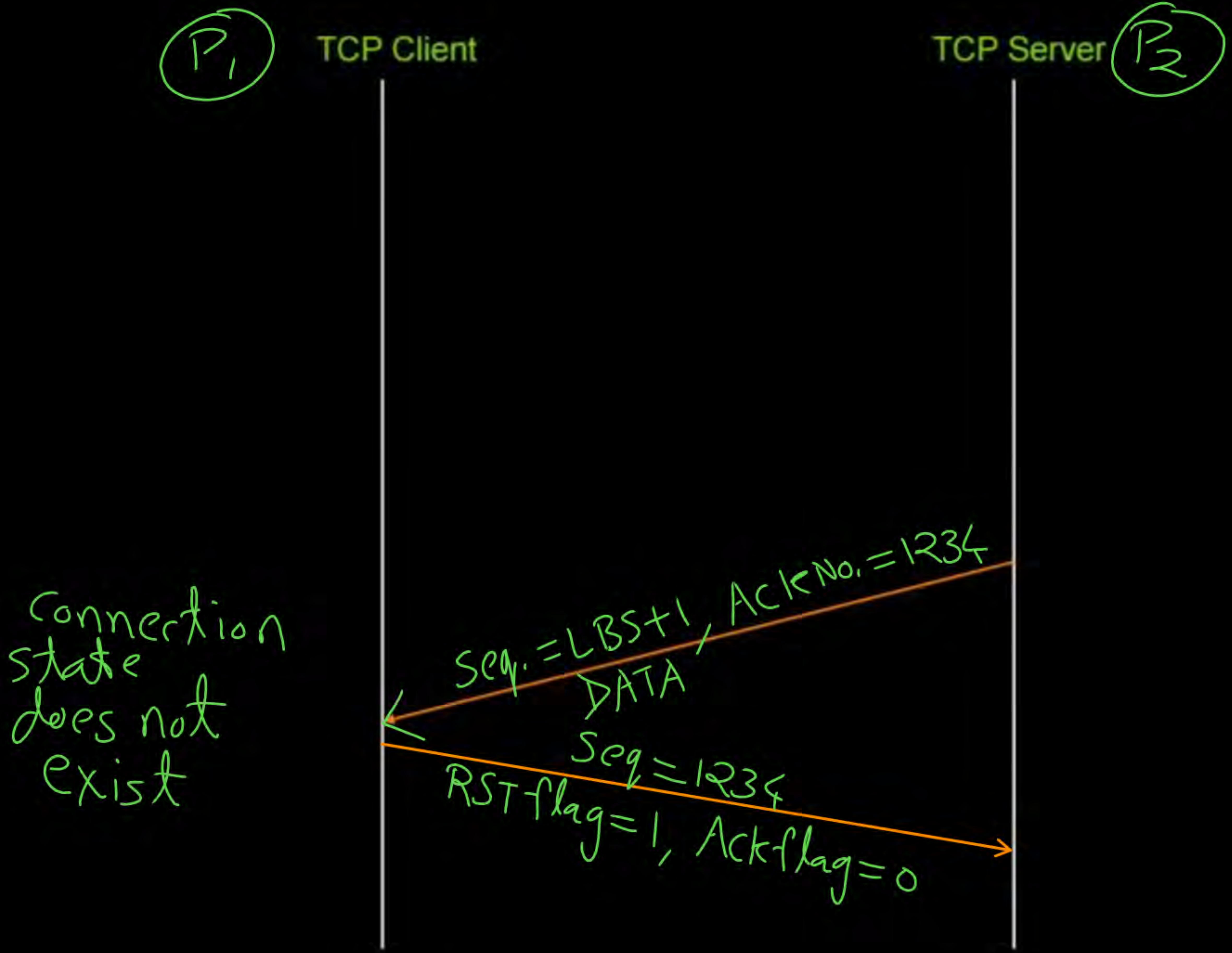
Topic : Reset Flag

→ TCP reset (RST) flag

→ Used to terminate TCP connection

→ When TCP host receives a TCP segment with the RST flag set to 1, it closes the connection


P1
TCP Client
TCP Server
P2
Data
Seq. No. = LBS+1
Ack. No. = 3123
Lost the connection state
or does not exist
Seq. No. = 3123
RST Flag=1, Ack flag=0


P1
TCP Client
TCP Server
P2
Seq.=LBS+1, AckNo.=1234
DATA
Connection state does not exist
Seq=1234
RST flag=1, Ackflag=0

Topic : TCP States

=> LISTEN :

→ The server is waiting for an incoming call

=> SYN_SENT :

→ The client has started to open a connection

=> SYN_RCVD :

→ A connection request has arrived, wait for ACK

=> ESTABLISHED :

→ Normal data transfer state

Topic : TCP States

=> FIN_WAIT-1 :

→ The client has said it is finished

=> CLOSE_WAIT :

→ The server has initiated a release

=> FIN_WAIT-2 :

→ The server has agreed to release

=> LAST_ACK :

→ Wait for pending packets

=> TIME_WAIT :

→ Wait for pending packets, "2MSL" Wait State

=> CLOSED :

→ No connection is active or pending

2 mins Summary

Topic TCP Connection Close

* MAC Sublayer
* Framing
* 2D parity, Hamming code

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* Supernetting, DHCP, NAT
* Application Layer

1. Bandwidth Delay Product (TCP Window Scaling)
2. TCP timers
3. Socket Programming
4. TCP Header options

THANK - YOU

@ABHISHEKSIRCS_PW